



A REPORT ON

UAC Care System Operations Dashboard

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Introduction:

The modern public service and healthcare ecosystem generates large volumes of operational data across multiple touchpoints, such as case intake, transfers, and discharges. Decision-makers often struggle to convert this raw data into timely, actionable insights, especially when teams rely on static spreadsheets or delayed reports. Interactive dashboards and analytical tools have therefore become essential for monitoring system load, understanding backlogs, and planning resources more effectively.

This project presents the development of an **operations command center dashboard**, built using Python and Streamlit, to visualize key indicators related to the UAC (Unaccompanied Alien Children) care system. The dashboard consolidates daily and monthly metrics from a SQLite database and presents them via intuitive views, including Executive Summary, Operations Monitor, Backlog & Risk, and Scenario Lab. By providing a single, integrated interface, the application helps users track total system load, intake pressure, discharges, and scenario-based projections.

From a management perspective, the dashboard demonstrates how data-driven tools can improve situational awareness, support proactive decision-making, and enhance accountability. It bridges the gap between operational data and strategic planning by transforming complex tables into clear visual stories that can be understood by both technical and non-technical stakeholders. The project shows how lightweight technologies like Streamlit and SQLite can be combined to deliver a practical decision-support solution.

Objectives:

The primary objective of this project is to design and implement a dashboard that supports monitoring and planning for the UAC care system in a clear, interactive way. The specific goals are:

1. **Provide an Executive Summary of system health** by surfacing headline indicators such as total system load, net intake pressure, 14-day backlog, and discharge offset ratio for the latest period.
2. **Monitor throughput and backlog over time** using daily, weekly, and monthly visualisations that track apprehensions, transfers, discharges, and system load.

3. **Support risk-aware decision-making** via a composite operational risk score and a Risk Driver Intensity chart that explains which factors contribute most to current risk.
 4. **Enable anomaly detection and short-term forecasting so that managers can detect unusual trends in system load and anticipate future pressure.**
 5. **Ensure data quality and transparency** by exposing row counts, date coverage, and summary statistics for all key data sources.
 6. **Demonstrate a reproducible data stack** using SQLite, Python, Streamlit, and Power BI connected through ODBC, suitable for extension or deployment in similar public-sector contexts.
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About the Dataset

The underlying dataset is stored in a SQLite database and is organised into multiple views that summarise daily and monthly activity in the UAC care system. These views are designed to capture intake, custody, transfers, and discharge information at different levels of aggregation to support both detailed and high-level analysis.

Key Data Components

1. Dashboard Feed View (vw_uac_dashboard_feed)

- Contains daily metrics such as report date, total system load, net intake pressure, intakes, transfers, discharges, and backlog measures.
- Supports the Executive Summary and Operations Monitor tabs by providing time-series data at the day level.

2. KPI Daily View (vw_uac_kpi_daily)

- Stores key performance indicators including CBP transfers, HHS discharges, and net intake pressure by day.
- Drives the Scenario Lab by supplying the recent 30-day pattern used as the baseline for what-if simulations.

3. Monthly Summary View (vw_uac_monthly_summary)

- Aggregates data by month, including average total system load, total net intake pressure, average discharge offset ratio, and related counts.
- Enables the dashboard to compare performance across months and years, and to estimate overall system pressure.

Other Data Components

1. Additional views such as `vw_uac_missing_dates`, `vw_uac_duplicate_dates`, and `vw_uac_rule_violations` support data quality diagnostics.
 2. **`vw_uac_weekly_summary` and `vw_uac_anomaly_summary`** – Derived views used in Power BI to summarise weekly flows and anomaly metrics.
 3. **`uac_raw`** – Raw feed of daily records capturing CBP apprehensions, transfers, HHS care counts, and discharges.
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Relevance of the Dataset to This Project

- The **daily views** support fine-grained monitoring of operational trends, which is crucial for detecting sudden spikes in intake or bottlenecks in discharges.
 - The **monthly summary** enables management to evaluate long-term patterns and to compare current performance against previous years.
 - The clear separation between feed, KPI, and monthly tables makes it easier to build focused visualisations for each dashboard tab without overloading users with raw data.
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Methodology

1. Data Preparation

The **first step** of the project involved preparing the raw UAC system data in a form that could be directly used for reporting and dashboard development. Instead of querying raw records repeatedly inside the dashboard, the data was organised into SQLite views so that the application could work with clean, analysis-ready tables. This reduced complexity at the front end and improved consistency between the Streamlit app and the Power BI report.

The SQLite database contained a base raw table and several derived views, each designed for a specific analytical purpose. For example, `vw_uac_dashboard_feed` served as the main daily reporting layer with variables such as `report_date`, `total_system_load`, `net_intake_pressure`, `backlog_accumulation_14d`, and `discharge_offset_ratio`. Similarly, `vw_uac_kpi_daily` contained operational measures such as CBP apprehensions, CBP transfers out, HHS discharges, HHS care counts, and seven-day rolling averages. The `vw_uac_monthly_summary` view aggregated records at the monthly level and included measures such as `avg_total_system_load`, `total_net_intake_pressure`, and `avg_discharge_offset_ratio`.

In addition to reporting views, supporting views were created for data quality assurance, such as missing dates, duplicate dates, rule violations, weekly summaries, and anomaly summaries. These views made it possible to identify gaps or inconsistencies before data was exposed in visualisations. As a result, the dashboard relied on a curated and validated data layer rather than directly consuming unstructured operational records.

2. Data Modelling and Integration

After the database views were prepared, the next step was to integrate them with the reporting tools. The project used an ODBC connection to SQLite, allowing Power BI to access the same database that was later used by the Python application. This established a single source of truth and ensured that both the Power BI dashboard and the Streamlit command center were based on the same definitions of KPIs and summary measures.

Within Power BI, the required views were selected from the ODBC source, including `vw_uac_dashboard_feed`, `vw_uac_kpi_daily`, `vw_uac_monthly_summary`, `vw_uac_weekly_summary`, and the data quality views. The fields were then reviewed in the model pane to confirm their types, hierarchies, and analytical roles. Date fields such as `report_date`, `month_start_date`, and `month_end_date` were used as temporal anchors, while numeric fields such as `total_system_load`, `cbp_apprehended`, and `hhs_discharged` were treated as measures.

This integration step was critical because it provided the foundation for consistent reporting across tools. By structuring the system around reusable SQL views, the project avoided duplication of transformation logic and made future maintenance easier.

3. Application Design

The dashboard design phase focused on converting operational and managerial requirements into an interface that would be easy to interpret. Rather than presenting all information on a single screen, the application was divided into multiple logical sections, with each tab representing a different decision context.

The Streamlit command center was organised into tabs such as Executive Summary, Operations Monitor, Backlog & Risk, Forecast & Anomalies, Scenario Lab, and Data Quality. The Executive Summary was designed for senior stakeholders who need quick visibility into the latest state of the system through KPI cards and a composite risk score. The Forecast & Anomalies tab supported forward-looking analysis, while the Data Quality tab ensured transparency regarding row counts, time coverage, and monthly summary records.

The filter panel was placed on the left side of the interface so that users could consistently control the year, trend frequency, date range, and primary metric without navigating away from the visualisations. This design allowed the same dashboard to serve both exploratory analysis and executive monitoring. The overall layout was kept clean and modular, ensuring that operational data could be consumed quickly without overwhelming the user.

Alongside Streamlit, the Power BI report was designed as a more detailed analytical workspace with multi-year comparisons, drill-down interactions, and dense visual composition. This complementary design strategy ensured that the project addressed both lightweight operational monitoring and deeper business intelligence exploration.

4. Implementation in Streamlit and Python

After the data model and interface structure were finalised, the application was implemented in Python using Streamlit. The Python script first connected to the SQLite database using the sqlite3 library and then loaded the required views into Pandas DataFrames. Date columns were explicitly converted into datetime format to support filtering, grouping, and time-series charting.

The application logic was written to calculate and display core KPIs such as total system load, net intake pressure, backlog accumulation, and discharge offset ratio. Plotly Express was used to create charts for system load trends, annual flow comparison, monthly summaries, anomaly detection, forecasting, and scenario analysis. Streamlit components such

as tabs, sliders, multiselects, metric, and plotly_chart were used to build an interactive interface without requiring a separate front-end framework.

Special attention was given to filter logic because the application needed to behave correctly across years and time frequencies. For example, year filters had to apply consistently to daily and monthly datasets, while scenario simulation required a valid subset of recent daily KPI records. Where necessary, additional guard conditions were introduced so that empty datasets would display warnings instead of blank charts.

The Forecast & Anomalies functionality was implemented using recent historical patterns of Total System Load. User-controlled sliders allowed changes in anomaly sensitivity, forecast horizon, and training period, making the analysis adjustable without changing code. Similarly, the Scenario Lab used sliders to model increases or decreases in CBP transfers out and HHS discharges, and then recalculated a simulated net intake pressure series. This interactive modelling approach made the dashboard more than just a reporting tool; it became a decision-support system.

5. Visual Design and Presentation Logic

The visualisation layer of the project was developed with the goal of balancing clarity and interpretability. KPI tiles were used for headline indicators because they communicate status quickly and are easy to compare. Bar charts and stacked bar charts were selected for yearly and weekly flow comparisons because they clearly show differences between apprehensions, transfers, discharges, and backlog levels.

Line charts were used for trend analysis, anomaly detection, and forecasts because they show temporal continuity and make directional changes more apparent. Horizontal bar charts were used for the Risk Driver Intensity visual so that the contribution of each operational factor could be understood immediately. The use of both summary tiles and explanatory charts ensured that the dashboard served users with different levels of analytical familiarity.

The Data Quality section was designed to add trust to the application by explicitly showing row counts and date coverage. This was important because users are more likely to rely on analytics when they can verify that the underlying data is complete and recent.

6. Testing and Iteration

Once the initial version of the dashboard was developed, a cycle of testing and refinement was carried out. Early iterations revealed issues such as blank tabs, mismatched column names, empty monthly filters, and scenario charts that did not populate properly when filtered data was insufficient. These problems were resolved by checking the structure of the SQLite views, validating the actual column names, computing missing year fields where necessary, and adding debugging outputs to inspect DataFrame content during execution.

Testing was performed across different years, filter selections, and slider settings to ensure that the application responded correctly under multiple conditions. For example, the monthly summary charts were checked to confirm that they rendered when valid monthly rows existed and showed a warning message when no rows were available. The scenario model was tested with different transfer and discharge assumptions to verify that the simulated net intake pressure changed logically.

The forecasting tab was also reviewed to confirm that anomaly sensitivity and forecast horizon controls affected the output as intended. On the presentation side, screenshots and dashboard layouts were reviewed to improve the readability of titles, card placement, and chart labelling. This iterative process ensured that the final system was not just functionally correct, but also understandable and presentation-ready for academic and managerial audiences.

7. Validation Approach

The project used a practical validation approach based on internal consistency, interpretability, and cross-tool agreement. Internal consistency was checked by comparing KPI values across views and ensuring that the same totals appeared in both Streamlit and Power BI outputs. Interpretability was assessed by verifying that charts told a coherent story—for example, a negative net intake pressure should align with higher discharge counts relative to apprehensions. Cross-tool agreement was confirmed by connecting both reporting environments to the same SQLite views and checking whether visuals reflected the same underlying data.

This validation strategy strengthened confidence in the analytical outputs and ensured that the dashboards could be used as a dependable decision-support tool.

8. Summary of Methodological Approach

In summary, the methodology combined data engineering, dashboard design, application programming, and iterative testing to create a robust UAC operations analytics platform. SQLite views provided a structured analytical backbone, Power BI enabled dense multi-year exploration, and Streamlit delivered an interactive and management-friendly command center. Through repeated refinement and validation, the project evolved from a raw data integration exercise into a fully functional dashboard system capable of supporting monitoring, forecasting, and scenario-based decision-making.

Features of the Operation Dashboard

The operations dashboard is designed as a **multi-tab web application** that combines data visualisation, KPIs, and lightweight simulation in a user-friendly interface. Its main features are:

1. Filters and Global Controls

On the left side of the Streamlit app, a filter panel allows users to choose:

- **Year selection** (individual year or “All years”).
- **Trend frequency** (daily, weekly, or monthly) to control how time series are aggregated.
- **Date range**, which restricts the analysis window across all tabs.
- **Primary metric**, used in some charts and anomaly detection (e.g., Total System Load).

A “Current Selection” section summarises how many daily records fall within the chosen filters, reinforcing transparency about the underlying data.

2. Executive Summary Tab

The Executive Summary tab provides a high level snapshot of system conditions.

- KPI tiles show Total System Load, Net Intake Pressure, 14 Day Backlog, and Discharge Offset Ratio, each compared with the previous value to show percentage change.
- A Composite Operational Risk gauge summarises multiple inputs into an index between 0 and 100, with a descriptive status such as “Stable”.

- The Risk Driver Intensity bar chart breaks down the composite risk into contributions from four drivers: system load, low discharge offset, backlog, and net pressure. In the example, system load contributes the majority of the risk score, while low discharge offset contributes a smaller share, and backlog/net pressure has a negligible contribution.

This tab is aimed at senior managers who need a quick picture of whether the system is approaching capacity or remains under control.

3. Forecast & Anomalies Tab

The Forecast & Anomalies tab introduces analytical capabilities beyond simple monitoring.

- Sliders let users adjust anomaly sensitivity, the baseline window (how many past days define “normal”), and the forecast horizon and training period for the predictive model.
- An Anomaly Detection chart plots Total System Load over time and highlights any days that fall outside the expected range. In the example screenshot, no anomalies are detected for the selected period, indicating a consistent pattern.
- A 30-Day Forecast chart shows the projected trajectory of Total System Load, with a purple confidence band around the forecasted line. Text panels below the chart state the forecast start and end dates and report the final projected load and its percentage change versus the latest actual.

These tools help planners anticipate rising pressure several weeks in advance and adjust transfers, discharges, or staffing strategies accordingly.

4. Scenario Lab Tab

The Scenario Lab supports “what if” exploration using recent KPI history.

- Two sliders allow the user to increase or decrease CBP transfers out and HHS discharges by chosen percentages relative to the baseline.
- The application applies these factors to the last 30 days of KPI data to compute a hypothetical net_intake_pressure_sim.

- A line chart then compares the baseline net intake pressure with the simulated scenario, showing how higher transfers or discharges could reduce future system load.

5. Interactive Filters

- A sidebar filter allows users to select one or more years, dynamically updating the charts and metrics across all tabs.
- This makes it easier to focus on a specific period, compare different years, or isolate recent performance for review.

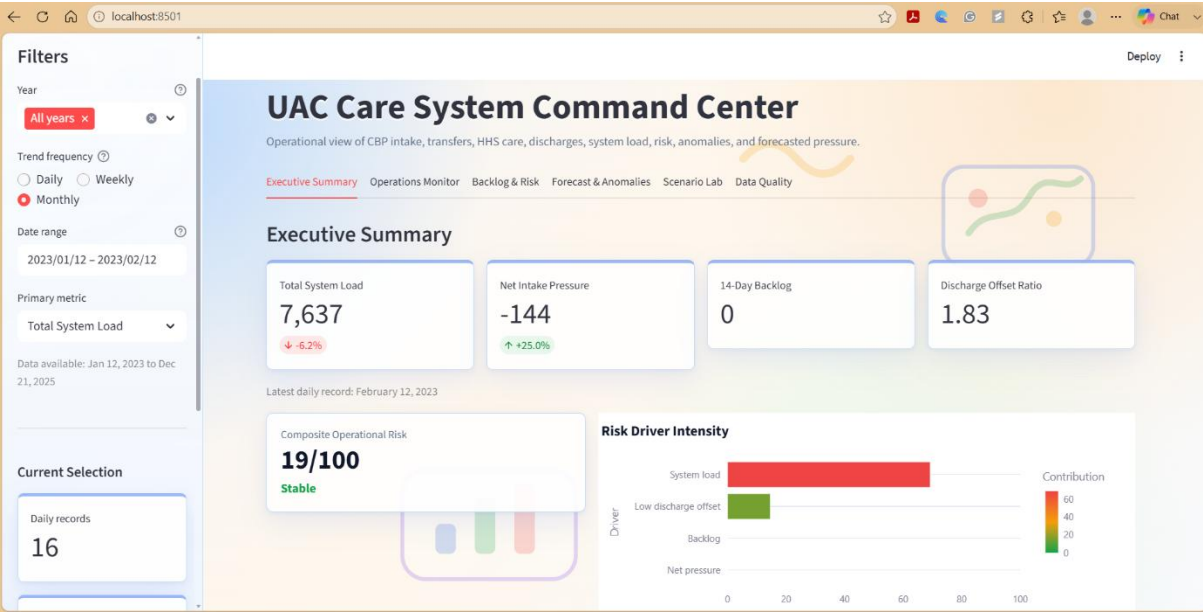
6. Data Quality Tab

The **Data Quality** tab documents the completeness and coverage of the underlying data.

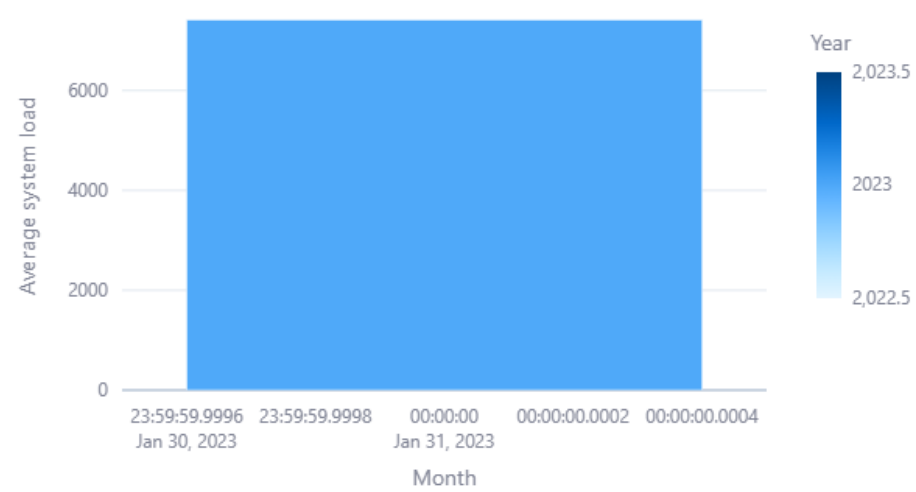
- Metric cards show the total number of rows in the **Dashboard Feed**, **KPI Daily**, and **Monthly Summary** views for the current filter.
- A table lists each view, the number of rows selected, and the start and end dates of those records.
- A **Monthly Summary Highlights** area shows, for the latest month, the average system load, net intake pressure, and discharge offset ratio as well as key counts.

This design ensures that users can quickly verify that their analyses are based on robust data, and it supports auditing and troubleshooting when something appears unusual in the visualisations.

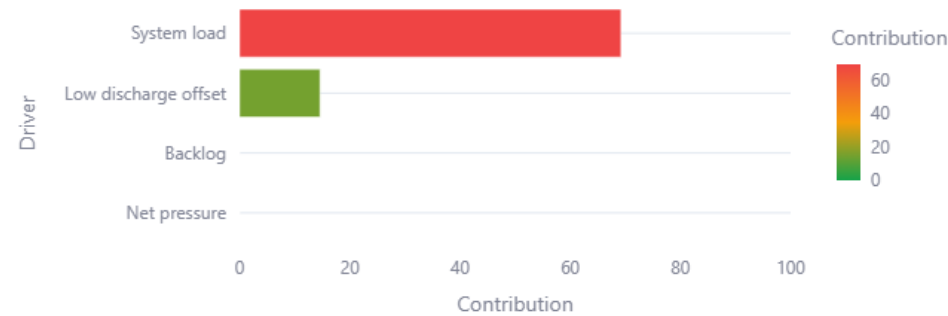
Images of the dashboard:

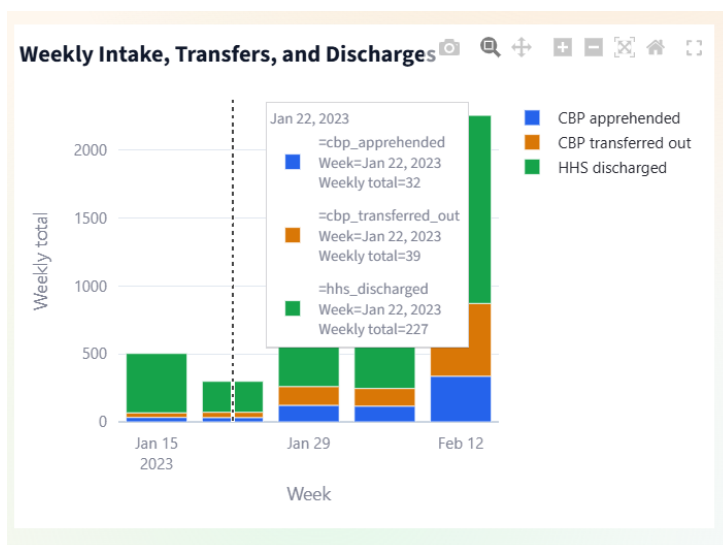
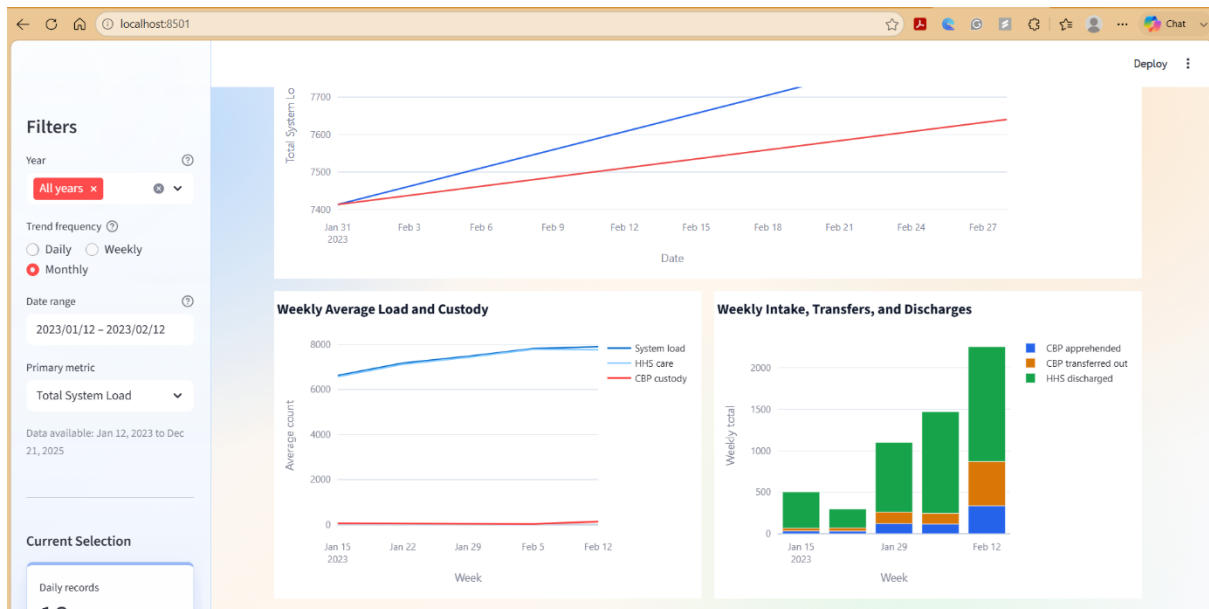
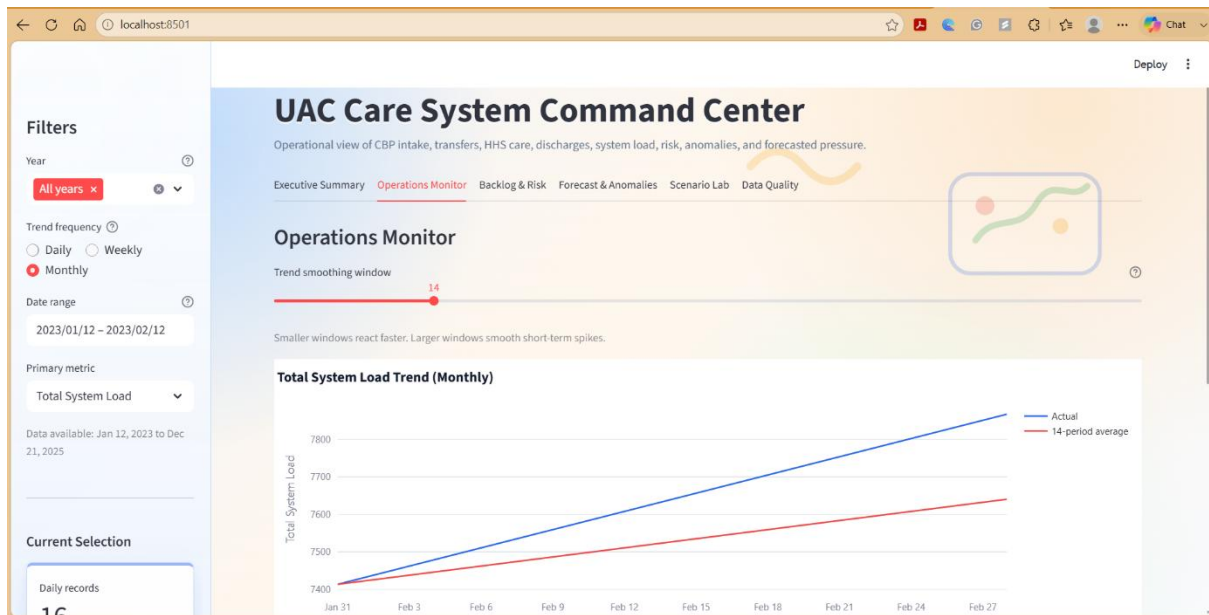


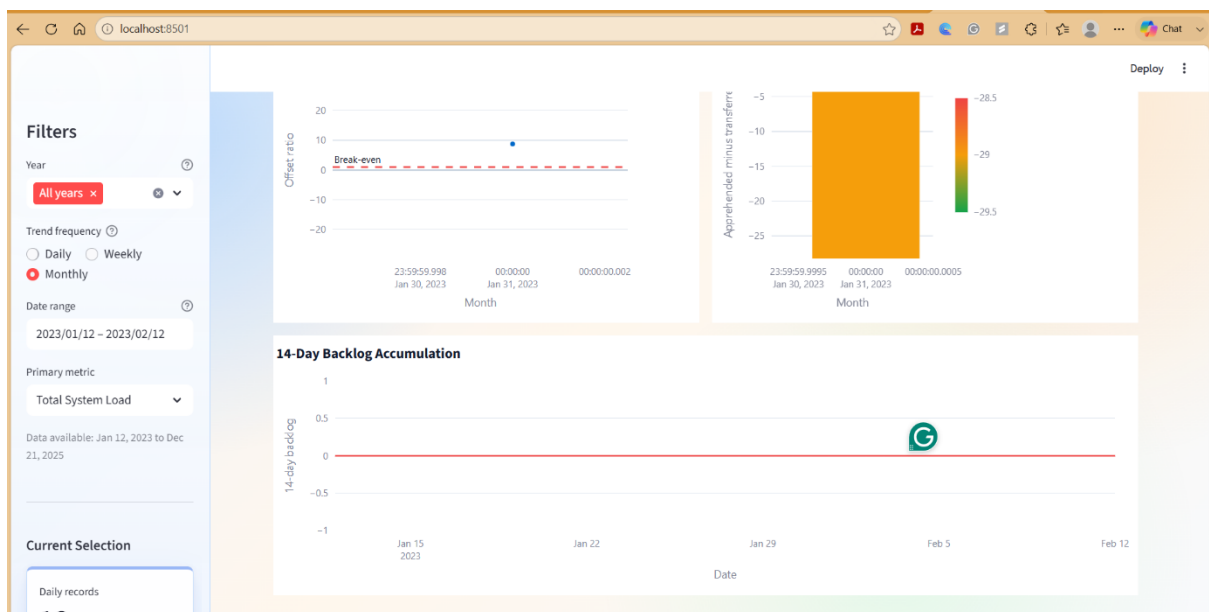
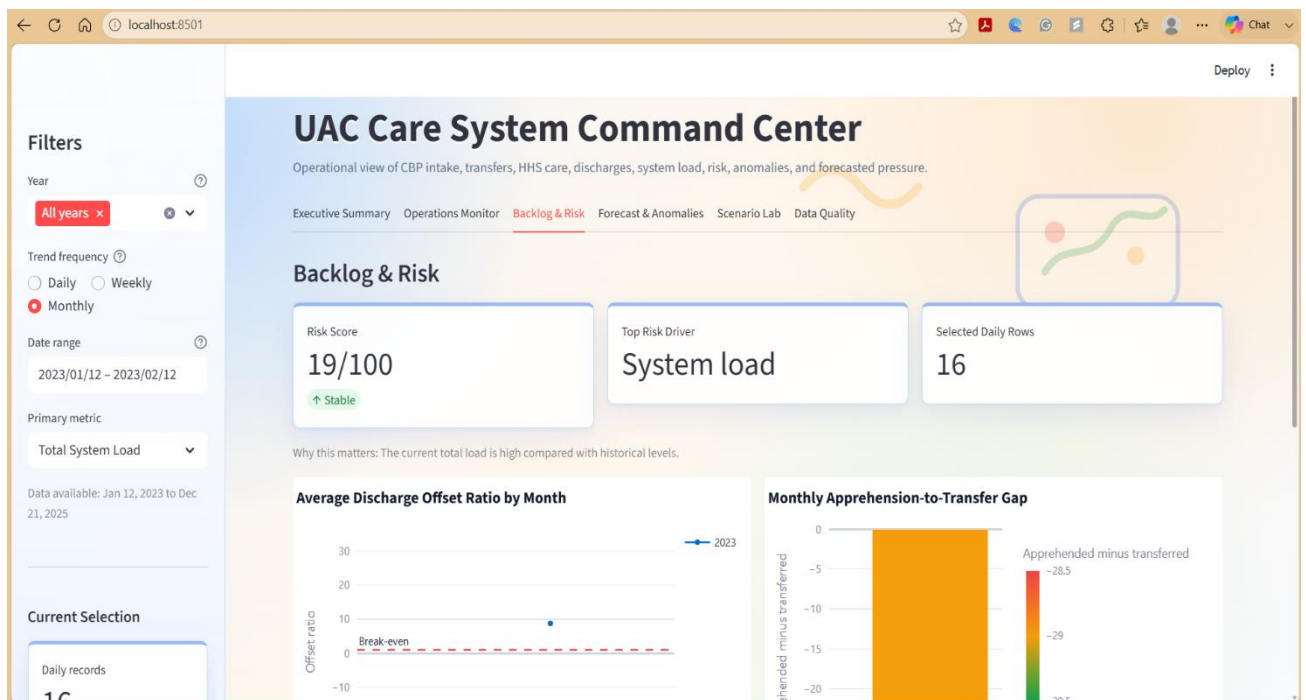
Average Monthly System Load

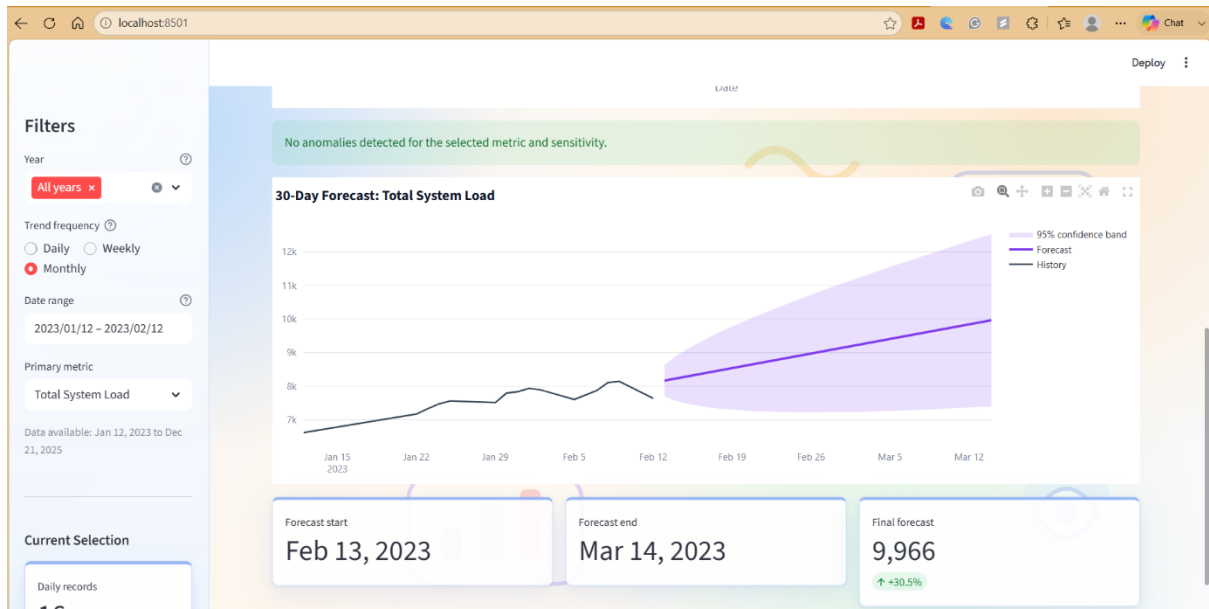
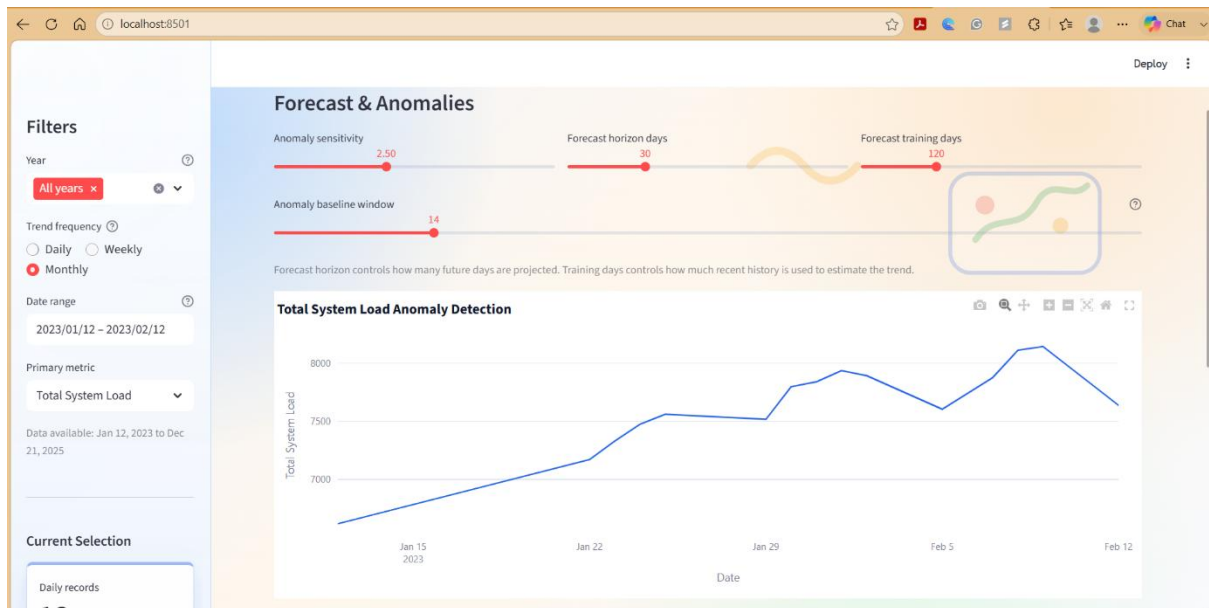


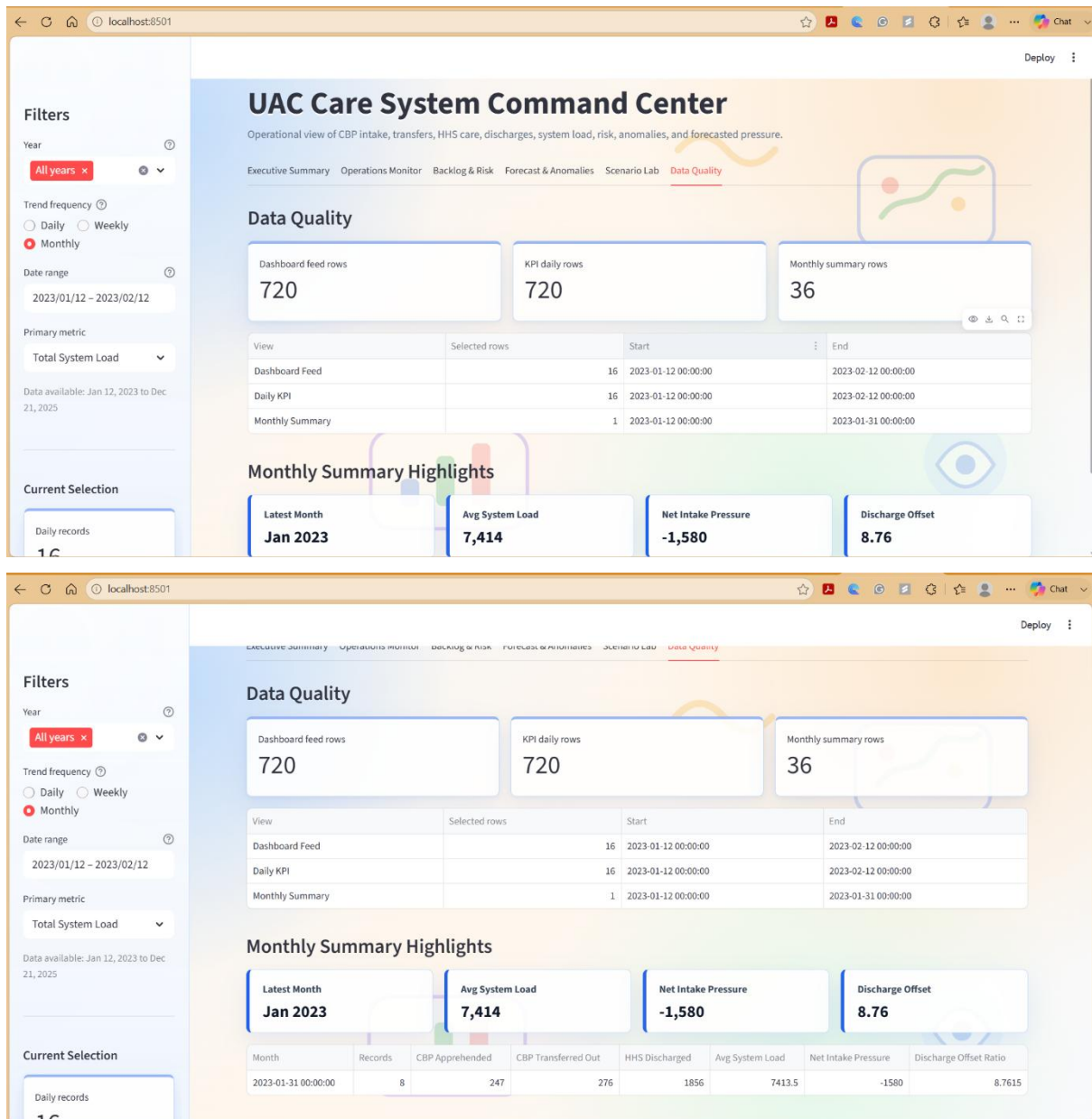
Risk Driver Intensity











Power BI Dashboard – System Load, Throughput, and Backlog

Parallel to the Streamlit application, a Power BI report provides a dense visual summary focused on multi-year comparison and drill-down analysis.

- Weekly Transfer-to-Apprehension and Discharge Offset Ratios**

A faceted bar chart displays Weekly Transfer-to-Apprehension and Discharge Offset Ratios by Year. For each week, side-by-side stacked bars show the proportion of CBP apprehensions that result in transfers versus the discharge offset ratio, across 2023, 2024, and 2025. This makes it

easy to see whether higher transfer efficiency is associated with higher discharge offsets and how these relationships evolve over time.

- **Annual System Load and Net Intake Pressure**

A pie chart labelled Annual System Load and Net Intake Pressure allocates total net intake pressure across years. In the snapshots, each of the years 2023, 2024, and 2025 accounts for roughly one-third of the total, but the chart allows for immediate detection of years where net pressure is unusually high.

- **Annual 14-Day Backlog Accumulation**

A horizontal bar chart shows Annual 14-Day Backlog Accumulation, comparing the volume of backlog days across years. For example, 2024 shows a backlog of around 114K, substantially higher than the 9K and 6K observed for 2025 and 2023, respectively. This signals that 2024 is the year where the system faced the greatest sustained pressure from backlogs, and it warrants deeper analysis.

- **Yearly CBP Apprehended vs. CBP Custody**

The Yearly CBP Apprehended vs. CBP Custody chart compares the number of children apprehended with the number in CBP custody at year-end. In the given views, custody counts (e.g., 46K and 70K) typically exceed annual apprehensions (27K and 37K), indicating that a significant portion of the caseload carries over from one year to the next. Drill-down capabilities allow users to explore this relationship by quarter.

- **Weekly System Load, Transfers, and Discharges**

A stacked column chart labelled Weekly System Load, Transfers, and Discharges presents the sum of total system load, total CBP transferred out, and total HHS discharged by week. The visual shows how weekly discharge volumes compare with intake and transfers, highlighting weeks where system load builds up or decreases. Combined with the backlog and ratio charts, this view helps explain *why* certain years or weeks experience rising or falling risk.

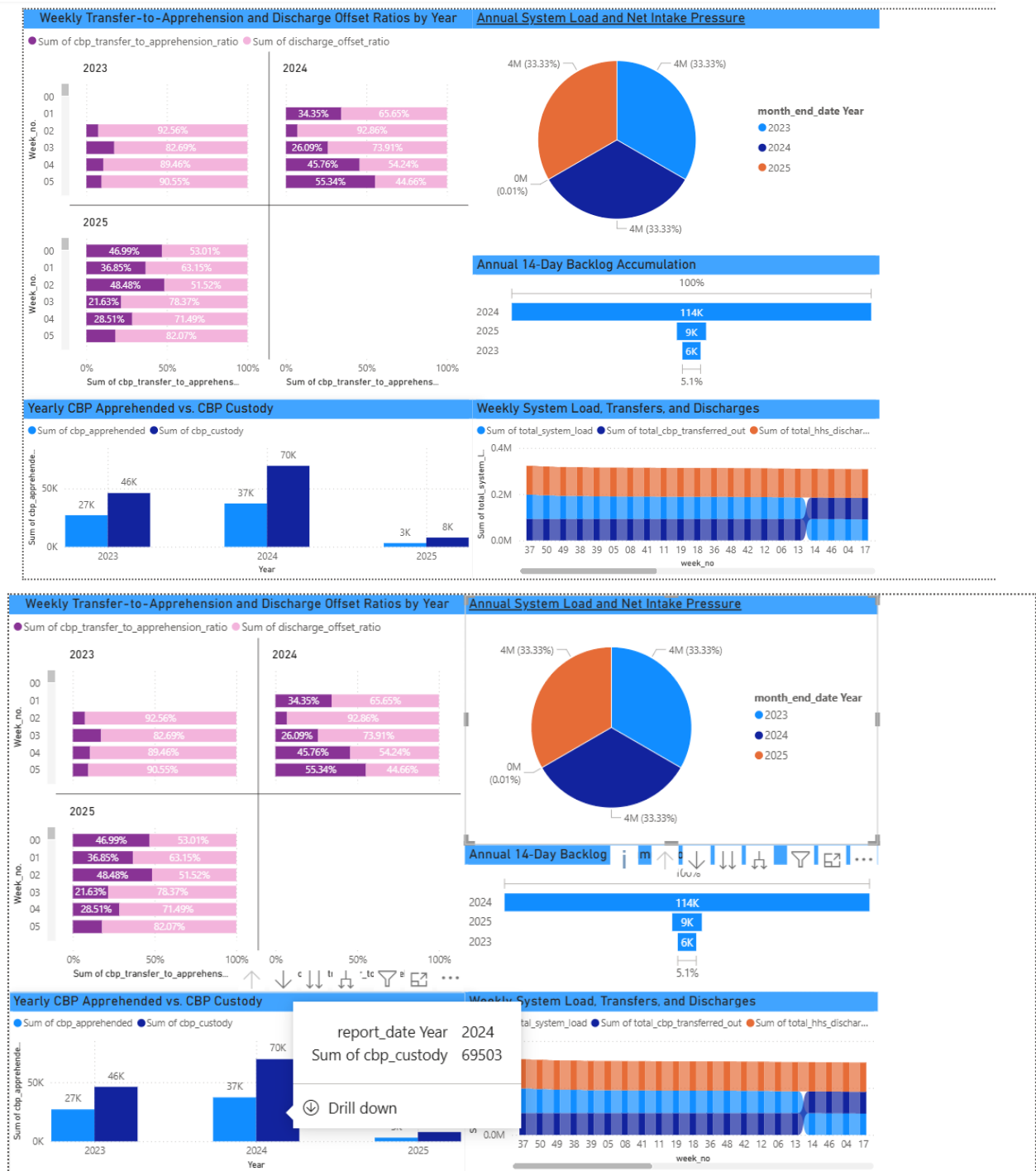
Power BI Visualizations:

The image displays three screenshots of the Power BI Data pane, illustrating different views of the data model. Each pane has a search bar at the top and a list of tables and columns below.

Top Left Pane: Shows a hierarchical view of the data model. The table `vw_uac_kpi_daily` is expanded, showing its columns: `avg_cbp_custody`, `avg_discharge_offset_ratio`, `avg_hhs_care`, `avg_total_system_load`, `month_end_date`, `month_start_date`, `record_count`, `total_cbp_apprehended`, `total_cbp_transferred_out`, `total_hhs_discharged`, and `total_net_intake_pressure`.

Top Right Pane: Shows a flat view of the data model. The table `vw_uac_kpi_daily` is expanded, showing its columns: `backlog_accumulation_14d`, `cbp_apprehended`, `cbp_custody`, `cbp_transfer_to_apprehensio...`, `cbp_transferred_out`, `discharge_offset_ratio`, `hhs_care`, `hhs_care_7d_avg`, `hhs_discharged`, `net_intake_pressure`, `net_intake_pressure_7d_avg`, `report_date`, `total_system_load`, and `total_system_load_7d_avg`.

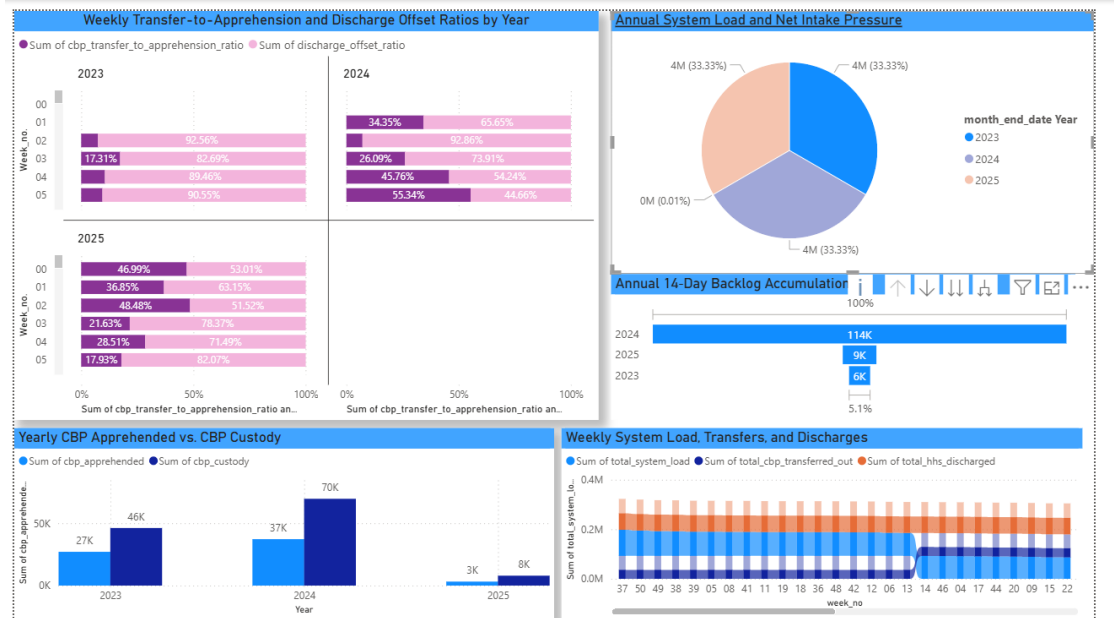
Bottom Left Pane: Shows a flat view of the data model. The table `vw_uac_kpi_daily` is expanded, showing its columns: `backlog_accumulation_14d`, `cbp_apprehended`, `cbp_custody`, `cbp_transfer_to_apprehensio...`, `cbp_transferred_out`, `discharge_offset_ratio`, `hhs_care`, `hhs_care_7d_avg`, `hhs_discharged`, `month`, `net_intake_pressure`, `net_intake_pressure_7d_avg`, `report_date`, `total_system_load`, `total_system_load_7d_avg`, `week_no`, and `year`.



The above figures gives a compact **multi-year overview** of system load, flows, and backlog.

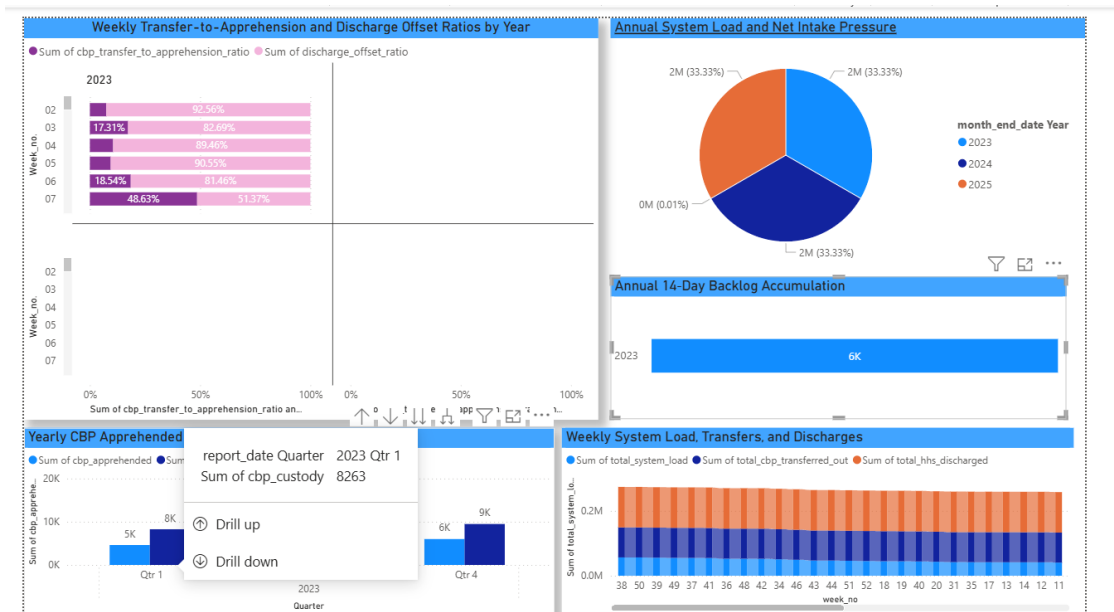
- Top left: Weekly transfer-to-arrest and discharge-offset ratios, by year, showing how efficiently cases move from CBP to HHS and out of the system.
- Top right: Pie chart of annual system load and net intake pressure, comparing how much pressure each year contributes.
- Middle right: Annual 14-day backlog accumulation, highlighting which year has the highest sustained backlog (2024 is clearly worst).

- Bottom left: Yearly CBP apprehended vs CBP custody counts, indicating how much of the caseload remains in CBP custody at year-end versus new apprehensions.
- Bottom right: Weekly system load, transfers, and discharges, showing how weekly flows of intakes and discharges drive total system load over time.



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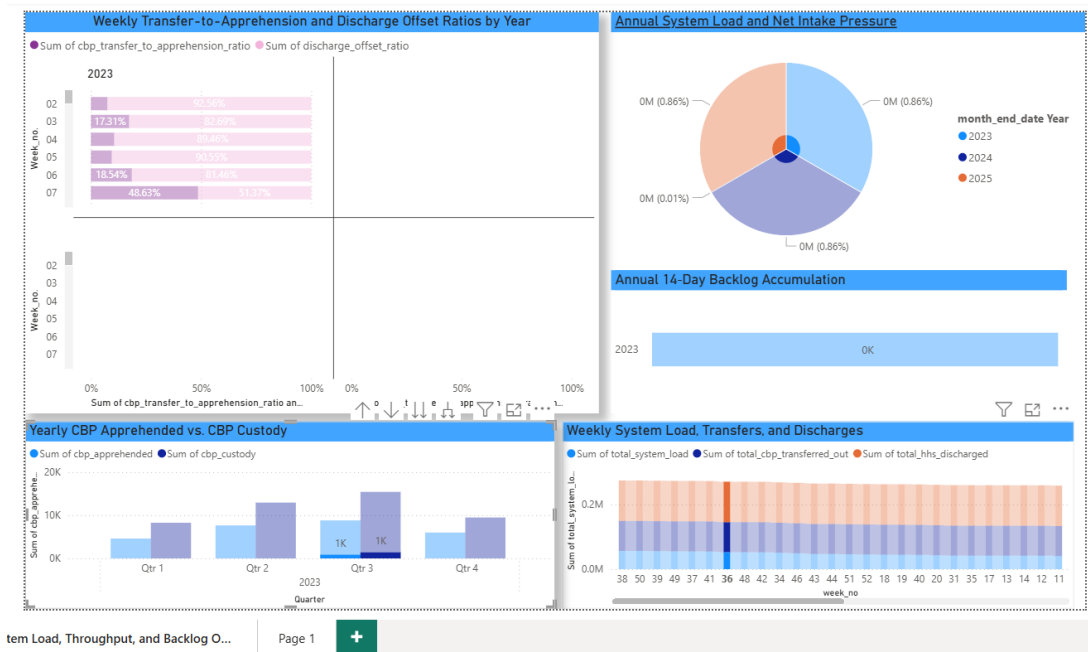
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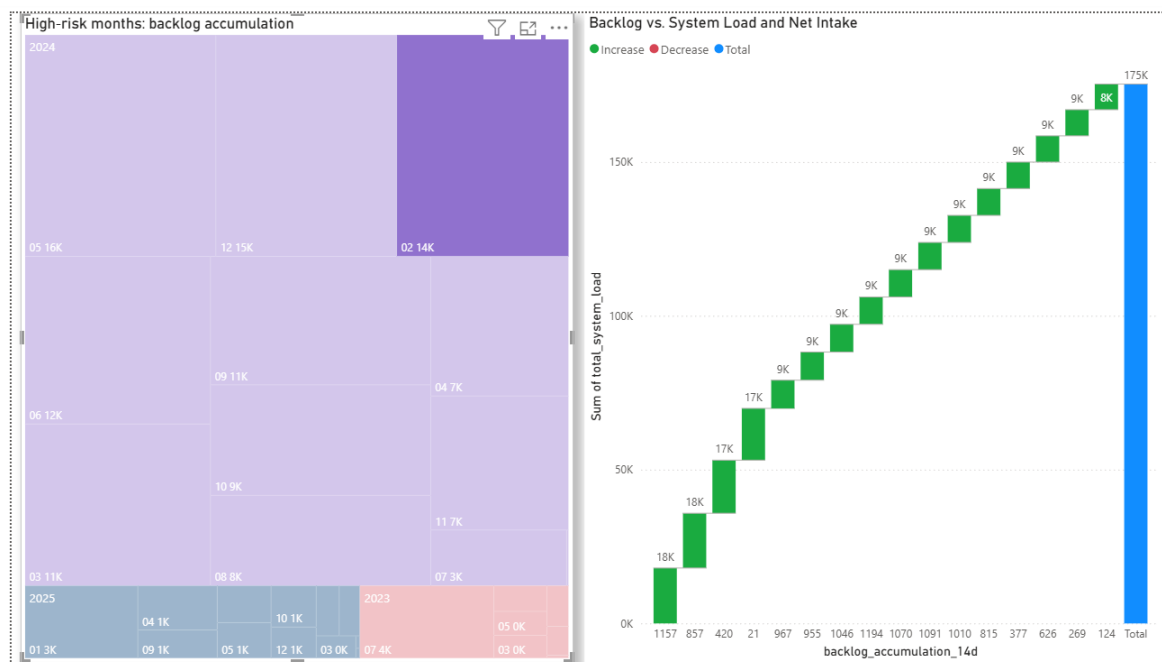
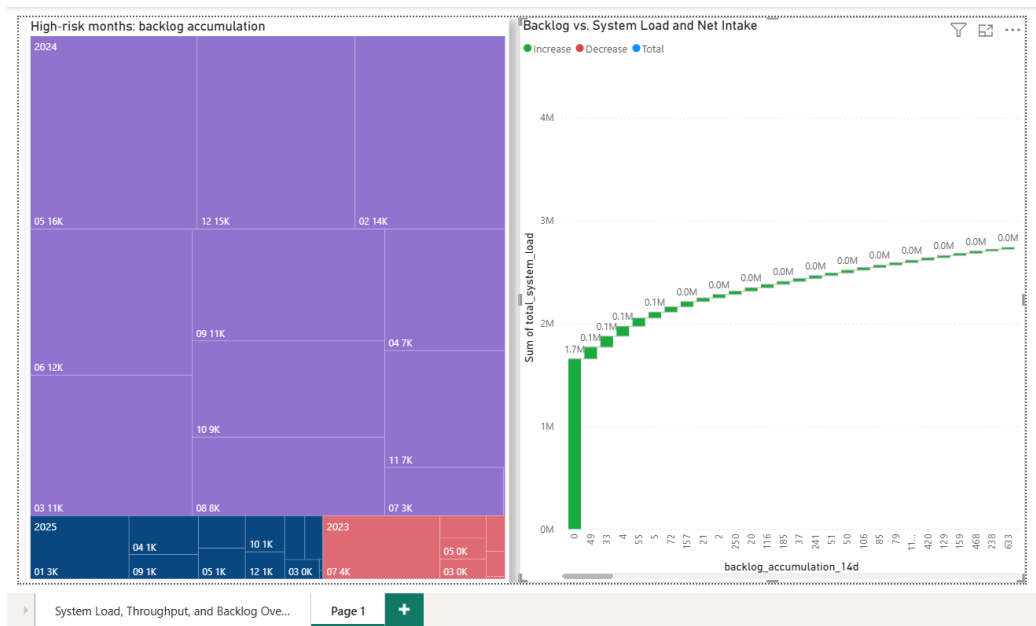




The above screenshot shows the same **System Load, Throughput, and Backlog** Power BI page as the overview, but with an important interactive feature highlighted — the **drill-down mode** is active on the *Yearly CBP Arrested vs. CBP Custody* chart.

Key points:

- **Top left** – Weekly Transfer-to-Arrest and Discharge Offset Ratios by year (2023, 2024, 2025), showing how efficiently CBP transfers children to HHS relative to total apprehensions each week.
- **Top right** – Pie chart of Annual System Load and Net Intake Pressure, showing each year's share of total system pressure across the full dataset.
- **Middle right** – Annual 14-Day Backlog Accumulation bar, now showing only **2023's backlog (6K)** because the drill-down has filtered the view to that specific year.
- **Bottom left** – The *Yearly CBP Arrested vs. CBP Custody* chart has been **drilled down to quarter level for 2023**, showing a tooltip that reads report_date.Quarter: 2023 Qtr 1, Sum of cbp_custody: 8,263. This confirms that Power BI's native drill-down capability is working, allowing users to move from yearly to quarterly granularity with a single click.
- **Bottom right** – Weekly System Load, Transfers, and Discharges stacked bar chart, unchanged from the overview, providing consistent context.





Treemap Analysis: High-Risk Months – Backlog Accumulation

The treemap visual highlights the distribution of backlog accumulation across different months and years. Each rectangle represents a month, with larger and darker purple blocks indicating higher backlog volumes.

Key Findings

- 2024 contributed the highest backlog accumulation, as shown by the dominance of large purple blocks across the treemap.
- Within 2024, months such as February, March, and April recorded the largest backlog volumes, indicating periods where incoming workload significantly exceeded processing capacity.
- 2025 shows comparatively smaller backlog values, suggesting improved workload management and reduced accumulation compared to the previous year.
- The lighter-colored sections represent months with lower backlog accumulation, indicating periods of relatively balanced system load and throughput.

The treemap visualization indicates that backlog accumulation was primarily concentrated in 2024, with several months contributing significantly more than others. Larger and darker purple blocks represent periods of higher backlog, suggesting operational bottlenecks where workload exceeded processing capacity. In contrast, backlog levels in 2025 were considerably lower, indicating improved system performance and workload management. These findings highlight the need for proactive resource planning during peak demand periods to prevent future backlog growth.

Results and Insights:

Across the combined dashboards, several insights emerge:

- In early 2023, the average monthly system load is around 7,414 children, but net intake pressure for that month is negative, meaning discharges exceeded new intakes.
- Annual comparison charts show that 2024 experiences the highest cumulative 14-day backlog accumulation, suggesting that despite high discharge volumes, intake pressures were even stronger that year.
- Risk Driver Intensity analyses indicate that **system load** and **low discharge offset** are the primary drivers of operational risk; backlog and net pressure become critical only when these two indicators deteriorate simultaneously.
- Short-term forecasts for Total System Load show potential increases of roughly 30% over a 30-day horizon under current trends, underlining the importance of proactive interventions when such forecasts are observed.

These findings reinforce the value of having both real-time dashboards and exploratory BI views built on the same data model.

Conclusion and Future Scope

The UAC Care System Command Center demonstrates how a relatively simple technology stack, SQLite, Python, Streamlit, and Power BI, can be combined to deliver a rich operational analytics solution for a complex public-sector system. The dashboards provide managers with a single, integrated view of system load, flows, backlogs, and risk, while also offering anomaly detection, forecasting, and scenario analysis.

Future enhancements could include:

- Adding **cohort and duration analysis** (e.g., average length of stay in care) to complement the existing load and throughput metrics.
- Introducing more sophisticated **forecasting models** and scenario simulations that account for policy or seasonal effects.
- Extending the data model to incorporate **geographic breakdowns**, enabling regional comparisons and targeted interventions.
- Integrating automated **alerts** based on anomaly detection and risk thresholds, so that stakeholders are notified when conditions deteriorate.

Overall, the project illustrates how well-designed dashboards can transform raw operational data into actionable insight, supporting more responsive and evidence-based management of the UAC care system.